The Price of a Place in Line

Elias Rook

Abstract: A developer with private value and reliability competes against a publicly known fallback. On a four-type domain, we derive the exact collateral required for efficient priority. It exceeds both pivotal types' expected gains from priority. A bound on the implementable priority gap delivers a sharp welfare-loss bound and an optimal lottery for a specified family of priors. The implementing deposit refunds more after failure than success.

1. Model

We narrow the two-developer environment by making developer 0's type (V, q) public, with $V > 0$ and $q \in (0, 1)$. Developer 1 privately observes $(v, p) \in \{a, b\} \times \{r, s\}$, where $0 < a < b$ and $0 < r < s < 1$. Both are risk neutral. Completion events are independent, probabilities cannot be manipulated, and project values are unverifiable.

The regulator chooses a possibly random ordering. The second project is attempted exactly when the first fails; exclusion is unavailable. Completion is publicly observed only when attempted. A successful first project receives its value, a successful second project receives δ times its value, and failure yields zero, where $0 < \delta < 1$.

After reporting but before the priority lottery, developer 1 escrows $e \le L$. At settlement its refund is between zero and the deposit. Its net payment therefore lies in $[0, L]$. Refunds may depend on reports, the lottery, and all observed outcomes. Developer 0 pays zero; no other transfers or escrow holding costs exist. Participation occurs after learning one's type and before reporting, with outside option zero. Individual rationality (IR) requires nonnegative expected utility at every type. Truthful reporting must be weakly dominant (DSIC).

Write

$$k(p) = 1 - \delta + \delta p, \qquad c = k(q), \qquad Q = qV, \qquad S(v, p) = \frac{pv}{k(p)}.$$

The welfare gain from putting developer 1 first is

$$\Delta(v, p) = cpv - Qk(p).$$

Assume

$$S(a, s) < Q/c < S(b, r), \qquad rb < sa. \tag{1}$$

Efficiency therefore gives developer 1 priority exactly when $v = b$, although the pivotal high-value type $h = (b, r)$ values priority less than $\ell = (a, s)$.

2. Main Result

Theorem. Define

$$K = \frac{c[s(1-r)a - r(1-s)b]}{s - r}, \qquad F = \frac{crs(b-a)}{s-r}.$$

Then $0 < F < K$ and $K > csa$.

Efficient truthful implementation with IR is possible if and only if $L \ge K$. Moreover, for any prior μ, every feasible DSIC mechanism has expected welfare loss

$$\mathbb{E}_\mu[W^* - W] \ge \min\{\mu_h \Delta_h, \mu_\ell(-\Delta_\ell)\}\left(1 - \frac{L}{K}\right)_+, \tag{2}$$

Here $z_+ = \max\{z, 0\}$; W averages welfare over execution and mechanism randomization, and W^* is its efficient counterpart.

The bound is attained, hence is the optimal welfare frontier, for the prior assigning probabilities π and $1-\pi$ to h and ℓ, respectively, and zero to the other types, provided

$$\pi\Delta_h \geq (1-\pi)(-\Delta_\ell).$$

An optimal mechanism gives high-value reports first priority and low-value reports first priority with probability $1-d$, where $d=\min\{1, L/K\}$. Only high-value reports escrow dK, paying dK after success and dF after failure, with the balance refunded.

3. Proof

Let x_z be the probability of first priority after report z, and let $T_z(p)$ be its expected net payment when true reliability is p. Conditional on a report, averaging over the lottery and developer 0's outcome makes T_z affine in p. Its counterfactual endpoints $T_z(0), T_z(1)$ belong to $[0, L]$, since every realized payment does. This argument requires no observation of unattempted projects.

A true type (v, p) reporting z receives expected utility

$$pv[\delta(1-q)+cx_z]-T_z(p).$$

Set $d_0 = x_h - x_\ell$ and $D = T_h - T_\ell$. The two cross-type IC constraints give

$$D(r) \leq crb\, d_0, \qquad D(s) \geq csa\, d_0.$$

Consequently,

$$L \geq D(1) = \frac{(1-r)D(s)-(1-s)D(r)}{s-r} \geq Kd_0. \tag{3}$$

Efficiency requires $d_0 = 1$, proving necessity.

For the construction, put $f(p) = F + (K-F)p$. Direct substitution gives

$$f(r) = crb, \qquad f(s) = csa, \qquad K - F = \frac{c(sa-rb)}{s-r} > 0.$$

Also $F > 0$ and $K - csa = c(1-s)(sa-rb)/(s-r) > 0$. Under the proposed mechanism, the utility difference between reporting value b and value a is

$$d[cpv - f(p)] = dcr(v-b), \qquad p = r,$$

$$d[cpv - f(p)] = dcs(v-a), \qquad p = s.$$

Thus all four types report value truthfully. Reliability reports are unused: truth is weakly optimal, but reliability is not separately identified. A free low-value contract ensures IR, and developer 0's payoff is nonnegative. Payments satisfy $0 \leq dF \leq dK \leq L$, so refunds respect escrow and regulator revenue is nonnegative in every outcome. Setting $d = 1$ proves sufficiency.

Let $A = \mu_h \Delta_h$ and $B = \mu_\ell(-\Delta_\ell)$. Losses at the pivotal types alone are

$$A(1-x_h) + Bx_\ell \geq \min\{A, B\}(1-d_0).$$

All other losses are nonnegative. Combining this inequality with (3) proves (2). For the specified two-point prior, $A \geq B$, and the proposed mechanism has $x_h = 1$, $x_\ell = 1-d$. Its loss is exactly $B(1-d)$, attaining (2). QED.

4. Economic Implications

For an electricity-grid connection, data-centre capacity slot, or other infrastructure right, the model describes a privately assessed project competing with an audited fallback. Private reliability raises required collateral even when efficient priority depends only on value. The minimum-escrow contract retains more upon success, making priority expensive for the reliable low-value type. This conclusion relies on immutable completion probabilities. Under the specified prior, scarce collateral optimally broadens access to priority through a lottery.

At $L = 0$, every type prefers more priority and payments vanish, so DSIC forces report-independent allocation. As $\delta \to 1$, scores converge to values while the collateral premium can persist. If instead both projects certainly complete and $a < V < b$, minimum collateral is a: low-value IC requires it, and a fixed first-priority fee a suffices.

Execution-contingent incentives and uncertain scheduling are established by Porter et al. (2008), Chen et al. (2010), and Conitzer and Vidali (2014). Burguet et al. (2012), Che et al. (2022), and Mylovanov and Zapechelnyuk (2017) study limited liability, deposits, and bounded verification penalties. Here the exact bounds concern escrow with binary outcomes that verify neither value nor probability. The literature check found no matching formula; publication-level novelty has not been established.

References

Burguet, R., J.-J. Ganuza, and E. Hauk. 2012. "Limited liability and mechanism design in procurement." *Games and Economic Behavior* 76(1): 15-25. doi:10.1016/j.geb.2012.04.004.

Che, X., T. Li, J. Lu, and X. Zheng. 2022. "Deposit Requirements in Auctions." *American Economic Journal: Microeconomics* 14(4): 465-493. doi:10.1257/mic.20200206.

Chen, J., L. Xu, and A. Whinston. 2010. "Managing Project Failure Risk Through Contingent Contracts in Procurement Auctions." *Decision Analysis* 7(1): 23-39. doi:10.1287/deca.1090.0155.

Conitzer, V., and A. Vidali. 2014. "Mechanism Design for Scheduling with Uncertain Execution Time." *Proceedings of AAAI* 28(1): 623-629. doi:10.1609/aaai.v28i1.8793.

Mylovanov, T., and A. Zapechelnyuk. 2017. "Optimal Allocation with Ex Post Verification and Limited Penalties." *American Economic Review* 107(9): 2666-2694. doi:10.1257/aer.20140494.

Porter, R., A. Ronen, Y. Shoham, and M. Tennenholtz. 2008. "Fault tolerant mechanism design." *Artificial Intelligence* 172(15): 1783-1799. doi:10.1016/j.artint.2008.06.004.